

Structural Break Crisis Inference Framework

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Abstract

This paper develops a single-axiom framework for crisis inference. The central axiom states that a crisis occurs if and only if it is the delayed manifestation of a structural break. From this axiom, the paper derives a formal theory of crisis ancestry, break sufficiency, structural stability, falsifiability, cascade formation, crisis classification, and statistical inference. The framework is designed for economics, finance, political science, geopolitics, international relations, warfare economics, warfare, systemic risk, data science, and artificial intelligence. It gives a unified language for detecting structural breaks, estimating crisis onset, forecasting crisis severity, classifying crisis type, and evaluating intervention policies. The paper includes mathematical definitions, theorems, proofs, statistical procedures, algorithms, model classes, AI/ML extensions, vector diagrams, tables, a bibliography, and a glossary.

The paper ends with “The End”

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1 Introduction

Crises are often described as sudden, exceptional, or spontaneous. In financial markets, a crash may seem to appear without warning. In political systems, regime collapse may appear abrupt. In geopolitics, war may seem to erupt from a small incident. In macroeconomics, recession may seem to begin after a visible shock. In warfare, the failure of a front, supply chain, command hierarchy, or deterrence regime may appear nonlinear and discontinuous.

The central thesis of this paper is that crisis is not spontaneous. A crisis is the delayed manifestation of a structural break. The break may be statistical, institutional, financial, strategic, geopolitical, technological, informational, logistical, or military. Once the break occurs, the system enters a transformed state. The crisis is the observable failure of the system after this transformation propagates through the relevant state space.

The paper develops a formal system around one axiom:

Crisis occurs if and only if it is the delayed manifestation of a structural break.

This statement is stronger than a one-way causation rule. It says both that every structural break produces a crisis and that every crisis has a structural break ancestor. The resulting framework is restrictive, falsifiable, and operational. It transforms crisis inference into five tasks:

detect the break, estimate its timing, estimate lead time, forecast severity, classify crisis type.

The framework is intentionally broad. A structural break may be a change in the mean of a time series, a volatility regime shift, a correlation breakdown, a collapse of institutional credibility, a change in military logistics, a deterrence failure, a technological disruption, a narrative reversal, or a resource constraint crossing a binding threshold. The same mathematical structure can represent a market crash, sovereign debt crisis, banking panic, civil conflict, interstate war, alliance breakdown, energy shock, cyber cascade, military defeat, or political legitimacy crisis.

2 Mathematical Environment

2.1 Probability space and filtration

Let

$$(\Omega, \mathcal{F}, \mathbb{P})$$

be a probability space. Let

$$\{\mathcal{F}_t : t \in \mathbb{T}\}$$

be a filtration representing the information available up to time t . The time index set $\mathbb{T}$ may be discrete, such as

$$\mathbb{T} = \{1, \dots, T\},$$

or continuous, such as

$$\mathbb{T} = [0, T].$$

Let

$$X_t : \Omega \rightarrow \mathcal{X}$$

be an observed stochastic process adapted to $\mathcal{F}_t$. The vector X_t may contain economic, financial, political, geopolitical, military, informational, technological, or environmental variables.

2.2 Structural state

Let (Θ, d_Θ) be a metric space of structural states. A structural state is denoted

$$\theta_t \in \Theta.$$

The data-generating law at time t is

$$X_t \sim P_{\theta_t}.$$

The structural state may include multiple components:

$$\theta_t = (\mu_t, \sigma_t^2, \Sigma_t, F_t, \rho_t, \lambda_t, \Gamma_t, N_t, R_t, \Pi_t),$$

where

- μ_t is a location or equilibrium component;
- σ_t^2 is a variance component;
- Σ_t is a covariance or network-correlation component;
- F_t is a distributional or tail component;
- ρ_t is a dependence or memory component;
- λ_t is an institutional rule component;
- Γ_t is a strategic interaction component;
- N_t is a network topology component;
- R_t is a resource constraint component;
- Π_t is a belief, narrative, or expectation component.

The interpretation of θ_t depends on domain. In finance, it may describe returns, volatility, correlations, liquidity, and tail risk. In macroeconomics, it may describe output, inflation, employment, credit, and fiscal constraints. In international relations, it may describe alliance structure, power distribution, deterrence credibility, and bargaining space. In warfare, it may describe force posture, supply lines, attrition rates, command integrity, intelligence quality, and weapons effectiveness.

3 Primitive Definitions

Definition 1 (Structural break). *A structural break occurs at time $\tau \in \mathbb{T}$ if the structural state is discontinuous at τ . Define*

$$B_\tau \iff d_\Theta(\theta_{\tau-}, \theta_{\tau+}) > 0,$$

where

$$\theta_{\tau-} := \lim_{s \uparrow \tau} \theta_s, \quad \theta_{\tau+} := \lim_{s \downarrow \tau} \theta_s.$$

The break magnitude is

$$\delta(\tau) = \frac{d_\Theta(\theta_{\tau-}, \theta_{\tau+})}{1 + d_\Theta(\theta_{\tau-}, 0)}.$$

Definition 2 (Systemic instability functional). *A systemic instability functional is a nonnegative adapted process*

$$L_t = L(X_t, \theta_t, \mathcal{F}_{t-}) \geq 0$$

measuring the distance of the system from a stability region. The crisis threshold is a constant $\omega > 0$.

Definition 3 (Crisis). *A crisis occurs on interval $I \subseteq \mathbb{T}$ if*

$$C(I) \iff \sup_{t \in I} L_t > \omega.$$

The severity of the crisis on I is

$$S(I) = \sup_{t \in I} \frac{L_t}{\omega}.$$

Therefore,

$$C(I) \iff S(I) > 1.$$

Definition 4 (Crisis onset). *A crisis onset occurs at time $c \in \mathbb{T}$ if*

$$O_c \iff L_c > \omega \quad \text{and} \quad L_s \leq \omega$$

for all $s < c$ sufficiently close to c .

Definition 5 (Break-response map). *Let $\mathcal{B}$ be the set of structural breaks. A break-response map is a measurable mapping*

$$\Phi : \mathcal{B} \rightarrow \mathbb{R}_+ \times \mathbb{R}_+ \times \mathcal{K}$$

such that

$$\Phi(B_\tau) = (\Delta_\tau, S_\tau, K_\tau),$$

where $\Delta_\tau > 0$ is the lead time from break to crisis onset, $S_\tau > 1$ is severity, and $K_\tau \in \mathcal{K}$ is crisis class.

4 The Single Axiom

Axiom 1 (Structural Break–Crisis Equivalence). *For every structural break B_τ , there exists a measurable break-response map Φ such that*

$$\boxed{B_\tau \iff O_{\tau+\Delta_\tau}.$$

Equivalently, for every interval $I \subseteq \mathbb{T}$,

$$\boxed{C(I) \iff \exists \tau : B_\tau \text{ and } \tau + \Delta_\tau \in I.}$$

The axiom states that crisis is exactly the delayed manifestation of structural break. It imposes two directions:

$$B_\tau \Rightarrow O_{\tau+\Delta_\tau}$$

and

$$O_c \Rightarrow \exists \tau \leq c : B_\tau, \quad c = \tau + \Delta_\tau.$$

Thus, the framework denies spontaneous crisis. It also denies harmless structural break. A true structural break has a crisis image, and a true crisis has a break ancestor.

5 Logical Consequences

Theorem 1 (Break sufficiency). *Every structural break produces a crisis onset after its lead time:*

$$B_\tau \Rightarrow O_{\tau+\Delta_\tau}.$$

Proof. By the Structural Break–Crisis Equivalence Axiom,

$$B_\tau \iff O_{\tau+\Delta_\tau}.$$

The implication from left to right gives

$$B_\tau \Rightarrow O_{\tau+\Delta_\tau}.$$

□

Theorem 2 (Crisis ancestry). *Every crisis has at least one structural break ancestor:*

$$C(I) \Rightarrow \exists \tau : B_\tau \text{ and } \tau + \Delta_\tau \in I.$$

Proof. By the interval form of the axiom,

$$C(I) \iff \exists \tau : B_\tau \text{ and } \tau + \Delta_\tau \in I.$$

The implication follows immediately. □

Theorem 3 (Structural stability). *An interval I is crisis-free if and only if no structural break maps into I :*

$$\neg C(I) \iff \neg \exists \tau : B_\tau \text{ and } \tau + \Delta_\tau \in I.$$

Proof. The interval form of the axiom gives

$$C(I) \iff \exists \tau : B_\tau \text{ and } \tau + \Delta_\tau \in I.$$

Negating both sides gives the result. $\square$

Theorem 4 (No spontaneous crisis). *If $C(I)$ occurs, then there exists a break time $\tau \leq \sup I$ such that B_τ .*

Proof. By crisis ancestry, there exists τ such that

$$B_\tau \text{ and } \tau + \Delta_\tau \in I.$$

Since $\Delta_\tau > 0$, $\tau < \tau + \Delta_\tau \leq \sup I$. Therefore $\tau \leq \sup I$. $\square$

Theorem 5 (Falsifiability). *The framework is falsified by either of the following observations:*

$$B_\tau \text{ and } \neg O_{\tau+\Delta_\tau},$$

or

$$C(I) \text{ and } \neg \exists \tau : B_\tau \text{ and } \tau + \Delta_\tau \in I.$$

Proof. The first observation violates the implication

$$B_\tau \Rightarrow O_{\tau+\Delta_\tau}.$$

The second violates the implication

$$C(I) \Rightarrow \exists \tau : B_\tau \text{ and } \tau + \Delta_\tau \in I.$$

Both implications are consequences of the axiom. Hence either observation refutes the system. $\square$

6 Crisis Counting and Overlap

Let

$$\mathcal{B}_T = \{\tau \leq T : B_\tau\}$$

be the set of break times up to T , and let

$$\mathcal{O}_T = \{c \leq T : O_c\}$$

be the set of crisis onsets up to T . Define

$$g : \mathcal{B}_T \rightarrow \mathcal{O}_T, \quad g(\tau) = \tau + \Delta_\tau.$$

Proposition 1 (Counting relation). *If g is injective, then*

$$|\mathcal{O}_T| = |\mathcal{B}_T|.$$

If g is not injective, then

$$|\mathcal{O}_T| \leq |\mathcal{B}_T|.$$

Proof. The map g sends every break time to its crisis onset. If g is injective, distinct breaks produce distinct crisis onsets, so the cardinalities coincide. If g is not injective, multiple breaks may map to the same onset, implying weak inequality. $\square$

Definition 6 (Compound crisis). *A compound crisis occurs when multiple structural breaks map into overlapping crisis intervals. If*

$$I_i = [\tau_i + \Delta_{\tau_i}, \tau_i + \Delta_{\tau_i} + h_i],$$

then the crisis is compound if

$$I_i \cap I_j \neq \emptyset$$

for at least one $i \neq j$.

Proposition 2 (Cascade amplification). *Suppose that after a break B_{τ_1} , the effective crisis threshold becomes*

$$\omega_t = \omega q_t, \quad 0 < q_t \leq 1.$$

If a second break B_{τ_2} occurs during the vulnerable region generated by B_{τ_1} , and $q_t < 1$, then the normalized severity of the second break is amplified:

$$\frac{L_t}{\omega_t} > \frac{L_t}{\omega}.$$

Proof. Since $\omega_t = \omega q_t$ and $0 < q_t < 1$,

$$\omega_t < \omega.$$

For any $L_t > 0$,

$$\frac{L_t}{\omega_t} = \frac{L_t}{\omega q_t} = \frac{1}{q_t} \frac{L_t}{\omega} > \frac{L_t}{\omega}.$$

Thus the same physical or institutional disturbance has higher normalized severity in the vulnerable post-break state. $\square$

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7 Taxonomy of Structural Breaks and Crisis Classes

A structural break may occur in different components of θ_t . Each component corresponds to a crisis class.

Table 1: Structural break taxonomy

Class	Structural object	Break type	Crisis interpretation
I	μ_t	Mean or equilibrium shift	Level crisis, output collapse, price dislocation, policy regime jump
II	σ_t^2	Variance shift	Volatility crisis, risk explosion, market instability
III	Σ_t	Covariance shift	Contagion, correlation breakdown, systemic financial crisis
IV	F_t	Distribution or tail shift	Extreme-event crisis, black-swan regime, fat-tail transition
V	ρ_t	Dependence or memory shift	Persistence crisis, unit-root transition, hysteresis
VI	λ_t	Institutional rule shift	Constitutional, regulatory, legitimacy, governance crisis
VII	Γ_t	Strategic interaction shift	Bargaining failure, deterrence collapse, war onset
VIII	N_t	Network topology shift	Infrastructure cascade, supply-chain collapse, cyber contagion
IX	Π_t	Belief or narrative shift	Panic, bank run, expectation reversal, mass mobilization
X	R_t	Resource constraint shift	Energy crisis, famine, liquidity crisis, ammunition exhaustion

8 Economics and Finance

8.1 Macroeconomic structure

Let

$$X_t = (y_t, \pi_t, u_t, i_t, b_t, q_t)$$

where y_t is output, π_t inflation, u_t unemployment, i_t the policy rate, b_t debt burden, and q_t real asset prices. A macroeconomic structural break occurs when

$$d_{\Theta}(\theta_{\tau-}, \theta_{\tau+}) > 0$$

for the macroeconomic law

$$X_t = A(\theta_t)X_{t-1} + \varepsilon_t.$$

A recessionary crisis may be defined by

$$L_t = w_y \max\{0, y^* - y_t\} + w_u \max\{0, u_t - u^*\} + w_{\pi} |\pi_t - \pi^*| + w_b \max\{0, b_t - b^*\}.$$

Then

$$C(I) \iff \sup_{t \in I} L_t > \omega.$$

8.2 Financial structure

Let

$$r_t \in \mathbb{R}^n$$

be a vector of asset returns. A financial break may occur in

$$\mathbb{E}[r_t], \quad \text{Cov}(r_t), \quad F_t, \quad \text{or liquidity depth.}$$

A portfolio crisis for holdings $w \in \mathbb{R}^n$ may be defined by

$$L_t = -\min\{w'r_t, 0\} + \alpha \text{VaR}_t(w) + \beta \text{CVaR}_t(w) + \gamma \ell_t^{-1},$$

where ℓ_t is market liquidity. A financial crisis occurs when

$$\sup_{t \in I} L_t > \omega.$$

8.3 Banking and liquidity

Let D_t be deposits, A_t liquid assets, E_t equity, and R_t reserve access. A banking instability functional can be written as

$$L_t = \eta_1 \max\{0, D_t - A_t - R_t\} + \eta_2 \max\{0, -E_t\} + \eta_3 \Delta D_t^-,$$

where

$$\Delta D_t^- = \max\{0, D_{t-1} - D_t\}.$$

A deposit-belief break in Π_t may map into a bank-run crisis.

9 Political Science, Geopolitics, and International Relations

9.1 Political stability

Let

$$X_t = (G_t, L_t^p, P_t, M_t, E_t)$$

where G_t is government capacity, L_t^p legitimacy, P_t protest intensity, M_t media fragmentation, and E_t elite cohesion. A political instability functional may be

$$L_t^{pol} = a_1 \max\{0, G^* - G_t\} + a_2 \max\{0, L^{p,*} - L_t^p\} + a_3 P_t + a_4 (1 - E_t).$$

A political crisis occurs when

$$\sup_{t \in I} L_t^{pol} > \omega_{pol}.$$

Institutional breaks occur when rules, enforcement capacity, elite coalitions, or legitimacy expectations shift discontinuously.

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9.2 Geopolitical order

Let

$$X_t = (P_t^1, \dots, P_t^m, A_t, D_t, T_t)$$

where P_t^i is the power of state i , A_t alliance structure, D_t deterrence credibility, and T_t trade interdependence. A geopolitical break may occur in the distribution of power, alliance topology, deterrence beliefs, or trade dependence.

A geopolitical crisis functional can be

$$L_t^{geo} = b_1 \|\Delta A_t\| + b_2 \max\{0, D^* - D_t\} + b_3 \text{PowerTransition}_t + b_4 \text{SanctionStress}_t.$$

War risk increases when deterrence credibility falls below a threshold while power transition pressure rises.

9.3 International relations and bargaining

Let two actors i and j bargain over a good of size V . Let p_t be the probability that i wins in war, $c_i, c_j > 0$ war costs, and $x_t \in [0, V]$ the division offered to i . The bargaining range exists when

$$p_t V - c_i \leq x_t \leq p_t V + c_j.$$

A bargaining crisis occurs when the bargaining range collapses:

$$p_t V - c_i > p_t V + c_j,$$

or more generally when asymmetric information, commitment problems, or indivisibilities destroy feasible settlement. A strategic structural break may be represented as

$$\Gamma_{\tau-} \neq \Gamma_{\tau+},$$

where Γ_t is the game form, payoff map, information structure, or commitment technology.

10 Warfare Economics and Warfare

10.1 Military production and logistics

Let

$$X_t = (Q_t, A_t, S_t, F_t^{fuel}, U_t, C_t)$$

where Q_t is military output, A_t ammunition stock, S_t supply-line integrity, F_t^{fuel} fuel stock, U_t unit readiness, and C_t command integrity. A warfare instability functional may be

$$L_t^{war} = c_1 \max\{0, A^* - A_t\} + c_2 \max\{0, F^* - F_t^{fuel}\} + c_3(1 - S_t) + c_4(1 - U_t) + c_5(1 - C_t).$$

A military crisis occurs when

$$\sup_{t \in I} L_t^{war} > \omega_{war}.$$

A logistics break may occur when

$$S_{\tau-} > S^* \quad \text{and} \quad S_{\tau+} < S^*.$$

Under the axiom, this break maps to a delayed operational crisis.

10.2 Attrition dynamics

Let R_t and B_t denote opposing force strengths. A Lanchester-type model is

$$\frac{dR_t}{dt} = -\alpha_t B_t, \quad \frac{dB_t}{dt} = -\beta_t R_t.$$

A structural break in effectiveness occurs when

$$(\alpha_{\tau-}, \beta_{\tau-}) \neq (\alpha_{\tau+}, \beta_{\tau+}).$$

The crisis functional for side R may be

$$L_t^R = \max\{0, R^* - R_t\} + \zeta \max\{0, \alpha_t - \alpha^*\} + \chi(1 - C_t).$$

A battlefield crisis occurs when force level, attrition disadvantage, or command integrity crosses the threshold.

11 Statistical Inference

The axiom concerns true structural breaks and true crises. Empirical inference concerns noisy observations. The central statistical task is to estimate the posterior distribution of

$$(\tau, \delta_\tau, \Delta_\tau, S_\tau, K_\tau)$$

given data

$$X_{1:t}.$$

11.1 Hypothesis testing

The structural null hypothesis is

$$H_0 : d_\Theta(\theta_{\tau-}, \theta_{\tau+}) = 0 \quad \text{for all } \tau \in [1, n].$$

The structural alternative is

$$H_1 : \exists \tau \in [1, n] \text{ such that } d_\Theta(\theta_{\tau-}, \theta_{\tau+}) > 0.$$

Let M_n be a break statistic. Reject H_0 when

$$M_n > c_\alpha.$$

Under the axiom, rejection of H_0 implies evidence of crisis onset at

$$\hat{c} = \hat{\tau} + \hat{\Delta}_{\hat{\tau}}.$$

11.2 CUSUM statistic

For a scalar process X_t , define

$$\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t.$$

The CUSUM statistic is

$$M_n = \max_{1 \leq k \leq n} \left| \frac{1}{\hat{\sigma}\sqrt{n}} \sum_{t=1}^k (X_t - \bar{X}_n) \right|.$$

A break estimate is

$$\hat{\tau} = \arg \max_{1 \leq k \leq n} \left| \sum_{t=1}^k (X_t - \bar{X}_n) \right|.$$

11.3 Likelihood ratio break test

Suppose the likelihood under no break is

$$\ell_0(\theta) = \sum_{t=1}^n \log f(X_t; \theta).$$

For a candidate break time τ , the broken likelihood is

$$\ell_1(\theta_1, \theta_2; \tau) = \sum_{t=1}^{\tau} \log f(X_t; \theta_1) + \sum_{t=\tau+1}^n \log f(X_t; \theta_2).$$

The quasi-likelihood ratio statistic is

$$QLR_n = \sup_{\tau \in \mathcal{T}_n} 2 \left[\ell_1(\hat{\theta}_1(\tau), \hat{\theta}_2(\tau); \tau) - \ell_0(\hat{\theta}) \right].$$

11.4 Posterior crisis probability

Under the axiom,

$$\mathbb{P}(C(I) \mid B_\tau) = \mathbf{1}\{\tau + \Delta_\tau \in I\}.$$

Therefore,

$$\mathbb{P}(C(I) \mid X_{1:t}) = \int \mathbf{1}\{\tau + \Delta_\tau \in I\} d\Pi(\tau, \Delta_\tau, \delta_\tau, K_\tau \mid X_{1:t}).$$

Uncertainty is inferential, not structural. The system is uncertain about whether and where a break occurred, not about whether a true break has a crisis image.

11.5 Lead-time estimation

Let

$$\Delta_\tau = c - \tau$$

be the lead time from structural break to crisis onset. Model Δ_τ using survival analysis:

$$h(s \mid Z_\tau) = h_0(s) \exp(\beta' Z_\tau),$$

where Z_τ includes break magnitude, break class, domain indicators, network centrality, leverage, institutional fragility, and military-logistical vulnerability.

The survival function is

$$S_\Delta(s \mid Z_\tau) = \exp \left(- \int_0^s h(u \mid Z_\tau) du \right).$$

The expected lead time is

$$\mathbb{E}[\Delta_\tau \mid Z_\tau] = \int_0^\infty S_\Delta(s \mid Z_\tau) ds.$$

11.6 Severity forecasting

Let severity satisfy

$$S_\tau > 1.$$

For exceedances above threshold u , define

$$Y_\tau = S_\tau - u \quad \text{conditional on } S_\tau > u.$$

A generalized Pareto model is

$$\mathbb{P}(Y_\tau \leq y \mid Y_\tau > 0) = 1 - \left(1 + \xi \frac{y}{\sigma_\tau}\right)^{-1/\xi},$$

where

$$\sigma_\tau = \sigma_0 + \gamma\delta_\tau + \eta'Z_\tau.$$

The parameter ξ controls tail thickness. If $\xi > 1/2$, the variance may be infinite. If $\xi > 1$, the mean may be infinite.

12 Algorithms

12.1 Algorithm 1: Structural Break Crisis Inference

Table 2: Sequential Structural Break Crisis Inference

Step	Operation
1	Observe data stream $X_{1:t}$.
2	Estimate structural state $\hat{\theta}_t$.
3	Compute break statistic $M_t = \sup_{\tau \leq t} \Lambda(\tau, t)$.
4	If $M_t \leq c_\alpha(t)$, retain H_0 and continue monitoring.
5	If $M_t > c_\alpha(t)$, estimate break time $\hat{\tau}$.
6	Estimate break magnitude $\hat{\delta}(\hat{\tau})$.
7	Classify break type $\hat{K}_{\hat{\tau}}$.
8	Estimate lead time $\hat{\Delta}_{\hat{\tau}}$.
9	Forecast crisis onset $\hat{c} = \hat{\tau} + \hat{\Delta}_{\hat{\tau}}$.
10	Forecast severity $\hat{S}_{\hat{\tau}}$.
11	Issue crisis inference tuple $(\hat{\tau}, \hat{\delta}, \hat{K}, \hat{\Delta}, \hat{c}, \hat{S})$.

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12.2 Algorithm 2: Bayesian Crisis Posterior

Table 3: Bayesian Posterior Crisis Computation

Step	Operation
1	Specify prior $\Pi_0(\tau, \Delta, \delta, K)$.
2	Compute likelihood $p(X_{1:t} \mid \tau, \Delta, \delta, K)$.
3	Update posterior: $d\Pi_t \propto p(X_{1:t} \mid \tau, \Delta, \delta, K) d\Pi_0.$
4	For target interval I , compute $\mathbb{P}(C(I) \mid X_{1:t}) = \int \mathbf{1}\{\tau + \Delta \in I\} d\Pi_t.$
5	Report posterior onset distribution, severity distribution, and crisis class probabilities.

12.3 Algorithm 3: AI/ML Crisis Learning

Table 4: Machine Learning Pipeline for Crisis Inference

Step	Operation
1	Construct rolling windows $W_t = (X_{t-m}, \dots, X_t)$.
2	Extract features: CUSUM, volatility, correlation, tail index, network centrality, sentiment, logistics, policy variables.
3	Train break detector $f_b(W_t) \in [0, 1]$.
4	Train crisis class classifier $f_K(W_t) \in \Delta(\mathcal{K})$.
5	Train lead-time model $f_\Delta(W_t)$.
6	Train severity model $f_S(W_t)$.
7	Calibrate probabilities using validation crises and false alarms.
8	Deploy sequentially with drift monitoring and human-auditable explanations.

13 Data Science and AI/ML Extensions

13.1 Feature construction

Let

$$Z_t = \phi(X_{t-m:t})$$

be a feature map from rolling historical windows to predictors. A useful feature vector may include

$$Z_t = (M_t, V_t, R_t, H_t, Q_t, N_t, P_t, W_t),$$

where M_t is a break statistic, V_t volatility, R_t correlation instability, H_t tail index estimate, Q_t liquidity stress, N_t network fragility, P_t political instability, and W_t warfare-logistics stress.

13.2 Supervised break detection

A supervised break detector estimates

$$f_b(Z_t) = \mathbb{P}(B_t = 1 \mid Z_t).$$

For logistic learning,

$$f_b(Z_t) = \frac{1}{1 + \exp(-\beta' Z_t)}.$$

For tree-based learning, f_b is a nonlinear partition of feature space. For neural learning,

$$f_b(Z_t) = \sigma(W_L \varphi_{L-1}(\cdots \varphi_1(W_1 Z_t))).$$

13.3 Multitask crisis network

A multitask model can jointly estimate break probability, class, lead time, and severity:

$$f(Z_t) = (f_b(Z_t), f_K(Z_t), f_\Delta(Z_t), f_S(Z_t)).$$

A joint loss is

$$\mathcal{L} = \lambda_b \mathcal{L}_b + \lambda_K \mathcal{L}_K + \lambda_\Delta \mathcal{L}_\Delta + \lambda_S \mathcal{L}_S,$$

where

$$\mathcal{L}_b = -y_b \log f_b - (1 - y_b) \log(1 - f_b),$$

$$\mathcal{L}_K = -\sum_{k \in K} y_k \log f_{K,k},$$

$$\mathcal{L}_\Delta = (\Delta - \hat{\Delta})^2,$$

and

$$\mathcal{L}_S = (S - \hat{S})^2.$$

13.4 Reinforcement learning for intervention

Let $a_t \in \mathcal{A}$ be an intervention: monetary policy, liquidity injection, diplomatic signal, military reinforcement, cyber defense, logistical rerouting, capital control, ceasefire proposal, or public communication.

Let the state be

$$s_t = (X_t, \hat{\theta}_t, \hat{\mathbb{P}}(C(I) \mid X_{1:t})).$$

A crisis-control policy is

$$\pi(a_t \mid s_t).$$

The reward may be

$$r_t = -\lambda_1 \mathbf{1}\{C_t = 1\} - \lambda_2 S_t - \lambda_3 \text{Cost}(a_t) - \lambda_4 \text{FalseAlarm}_t.$$

The objective is

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t \right].$$

The policy does not invalidate the axiom. Intervention may change the structural state and thereby create a new break-response map. A successful intervention is itself a structural transformation that redirects the crisis image.

14 Model Classes

Table 5: Model families for structural break crisis inference

Model	Mathematical form	Application
SB-AR	$X_t = a_t + \rho_t X_{t-1} + \varepsilon_t$	Persistence and memory crises
SB-VAR	$X_t = A_t X_{t-1} + \varepsilon_t$	Macroeconomic, geopolitical, and systemic interactions
SB-GARCH	$\sigma_t^2 = \omega_t + \alpha_t \varepsilon_{t-1}^2 + \beta_t \sigma_{t-1}^2$	Volatility crises and financial instability
SB-DCC	$H_t = D_t R_t D_t$ with broken R_t	Correlation crises and contagion
SB-EVT	$S - u \sim GPD(\sigma_t, \xi_t)$	Tail crises and extreme events
SB-State Space	$x_t = A_t x_{t-1} + u_t, y_t = C_t x_t + v_t$	Latent breaks, hidden political or military stress
SB-Network	$A_t^{net} \neq A_{t-1}^{net}$	Supply-chain, alliance, cyber, infrastructure crises
SB-Game	$\Gamma_{\tau-} \neq \Gamma_{\tau+}$	Bargaining failure, deterrence collapse, war onset
SB-Agent	Agents update rules after τ	Panic, bank runs, mobilization, behavioral cascades
SB-ML	$f(Z_t) \rightarrow (\hat{B}, \hat{K}, \hat{\Delta}, \hat{S})$	High-dimensional early warning systems

15 Vector Graphics

15.1 Axiomatic flow

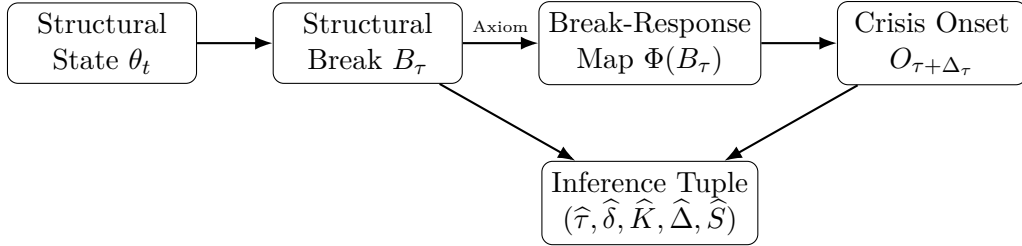


Figure 1: Structural break crisis inference as an axiomatic flow.

15.2 Crisis taxonomy

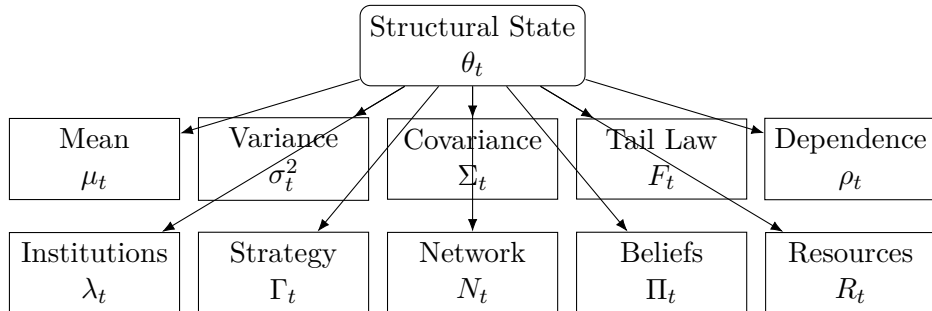


Figure 2: Structural state decomposition into crisis-generating components.

15.3 Sequential inference pipeline

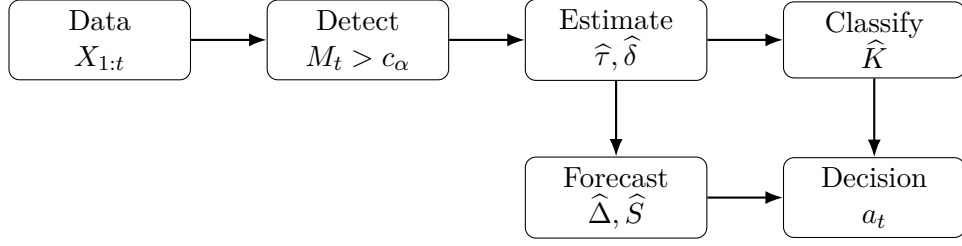


Figure 3: Operational pipeline for crisis inference and intervention.

16 Empirical Design

16.1 Event construction

An empirical study requires three datasets:

1. A panel of observed variables $X_{i,t}$ for unit i at time t .
2. A catalog of crisis onsets $O_{i,c}$.
3. A candidate set of break times $\hat{B}_{i,\tau}$.

For each unit i , the break-to-crisis sample consists of pairs

$$(\hat{\tau}_i, c_i), \quad \hat{\Delta}_i = c_i - \hat{\tau}_i.$$

Right censoring occurs if a break is detected but no crisis has yet appeared within the observation window.

16.2 Validation metrics

Detection quality can be measured by

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN},$$

and

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Lead-time accuracy can be measured by

$$MAE_{\Delta} = \frac{1}{N} \sum_{i=1}^N |\hat{\Delta}_i - \Delta_i|.$$

Severity accuracy can be measured by

$$RMSE_S = \left[\frac{1}{N} \sum_{i=1}^N (\hat{S}_i - S_i)^2 \right]^{1/2}.$$

16.3 False alarms and missed crises

A false alarm occurs when the model estimates a structural break and crisis image, but no crisis occurs within the admissible horizon. A missed crisis occurs when a crisis occurs but no break ancestor is detected.

The cost function is

$$\mathcal{C} = c_{FP}FP + c_{FN}FN + c_L|\widehat{\Delta} - \Delta| + c_S|\widehat{S} - S|.$$

In high-stakes domains such as banking crises, war onset, nuclear escalation, famine, or infrastructure failure, typically

$$c_{FN} \gg c_{FP}.$$

17 Intervention and Counterfactuals

The axiom states that a true break has a crisis image under the prevailing structure. However, intervention may alter the structure. Let a_t be an intervention at time t . The post-intervention structural state becomes

$$\theta_t^a = \Psi(\theta_t, a_t).$$

The post-intervention crisis image is

$$\Phi^a(B_\tau) = (\Delta_\tau^a, S_\tau^a, K_\tau^a).$$

An intervention is crisis-reducing if

$$S_\tau^a < S_\tau$$

or

$$\Delta_\tau^a > \Delta_\tau,$$

or if the crisis class is transformed into a less destructive class. A successful intervention does not contradict the axiom. It changes the structural mapping that governs the break-response relation.

18 Limitations

The framework is intentionally strict. Its strength is also its vulnerability. It can fail in four ways:

1. A crisis is observed without any identifiable structural break.
2. A structural break is observed without any crisis image.
3. The measured break is an artifact of noise or model misspecification.
4. The crisis threshold ω is poorly chosen.

The framework therefore requires careful measurement of θ_t , transparent crisis definitions, robust break detection, and domain-specific validation.

19 Conclusion

This paper has developed a single-axiom theory of crisis inference. The axiom states that crisis occurs if and only if it is the delayed manifestation of a structural break. From this axiom follow break sufficiency, crisis ancestry, structural stability, no spontaneous crisis, falsifiability, crisis counting, and compound crisis logic.

The framework converts crisis analysis into structural inference. The empirical problem is not whether breaks matter. The empirical problem is whether a break occurred, where it occurred, what kind it was, when its crisis image will arrive, and how severe that crisis will be. This architecture unifies statistical break testing, macro-financial crisis detection, political instability analysis, geopolitical risk, international bargaining failure, warfare logistics, AI/ML forecasting, and intervention design.

A Additional Mathematical Details

A.1 Consistency of break-time estimation

Assumption 1 (Identifiability). *There exists a unique break time τ_0 such that*

$$d_{\Theta}(\theta_{\tau_0^-}, \theta_{\tau_0^+}) > 0,$$

and for all $\tau \neq \tau_0$,

$$d_{\Theta}(\theta_{\tau^-}, \theta_{\tau^+}) = 0.$$

Assumption 2 (Signal domination). *The break statistic satisfies*

$$\mathbb{E}[M_n(\tau_0)] - \sup_{\tau \neq \tau_0} \mathbb{E}[M_n(\tau)] \rightarrow \infty.$$

Theorem 6 (Break-time consistency). *Under identifiability and signal domination,*

$$\hat{\tau} = \arg \max_{\tau \in \mathcal{T}_n} M_n(\tau)$$

satisfies

$$\hat{\tau} \xrightarrow{p} \tau_0.$$

Proof. For every $\varepsilon > 0$, define the complement neighborhood

$$A_{\varepsilon} = \{\tau : |\tau - \tau_0| > \varepsilon\}.$$

By signal domination,

$$\mathbb{P} \left(\sup_{\tau \in A_{\varepsilon}} M_n(\tau) \geq M_n(\tau_0) \right) \rightarrow 0.$$

Therefore,

$$\mathbb{P}(|\hat{\tau} - \tau_0| > \varepsilon) \rightarrow 0.$$

Hence $\hat{\tau} \xrightarrow{p} \tau_0$. □

A.2 Posterior concentration

Theorem 7 (Posterior concentration around crisis onset). *Suppose the posterior over (τ, Δ) satisfies*

$$\Pi_t(|(\tau + \Delta) - c_0| > \varepsilon \mid X_{1:t}) \rightarrow 0$$

in probability. Then

$$\mathbb{P}(C(I) \mid X_{1:t}) \rightarrow \mathbf{1}\{c_0 \in I\}$$

for every interval I whose boundary does not contain c_0 .

Proof. By the posterior crisis formula,

$$\mathbb{P}(C(I) \mid X_{1:t}) = \int \mathbf{1}\{\tau + \Delta \in I\} d\Pi_t.$$

If the posterior mass of $\tau + \Delta$ concentrates at c_0 , then the integral converges to $\mathbf{1}\{c_0 \in I\}$, provided c_0 is not on the boundary of I . $\square$

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Glossary

Structural break

A discontinuity in the structural state θ_t of the system.

Crisis

A threshold violation of a systemic instability functional.

Crisis onset

The first local time at which instability exceeds the crisis threshold.

Break-response map

The function mapping a structural break into lead time, severity, and crisis class.

Lead time

The delay Δ_τ between break time and crisis onset.

Severity

The normalized maximum instability produced by a crisis.

Crisis class

A qualitative type of crisis determined by the structural component that breaks.

Mean break

A shift in the equilibrium level of a process.

Variance break

A shift in volatility or uncertainty.

Covariance break

A shift in correlation, contagion, or network dependence.

Tail break

A shift in extreme-event probability.

Dependence break

A shift in memory, autocorrelation, or persistence.

Institutional break

A discontinuity in rules, legitimacy, enforcement, or governance.

Strategic break

A discontinuity in game form, bargaining space, or deterrence credibility.

Network break

A discontinuity in graph topology, connectivity, or centrality.

Belief break

A discontinuity in expectations, narratives, credibility, or mass psychology.

Resource break

A discontinuity in binding constraints such as liquidity, food, energy, ammunition, or logistics.

False alarm

A detected break-crisis signal without realized crisis within the admissible horizon.

Missed crisis

A realized crisis whose structural break ancestor was not detected.

Cascade

A compound crisis generated by overlapping or sequential structural breaks.

The End